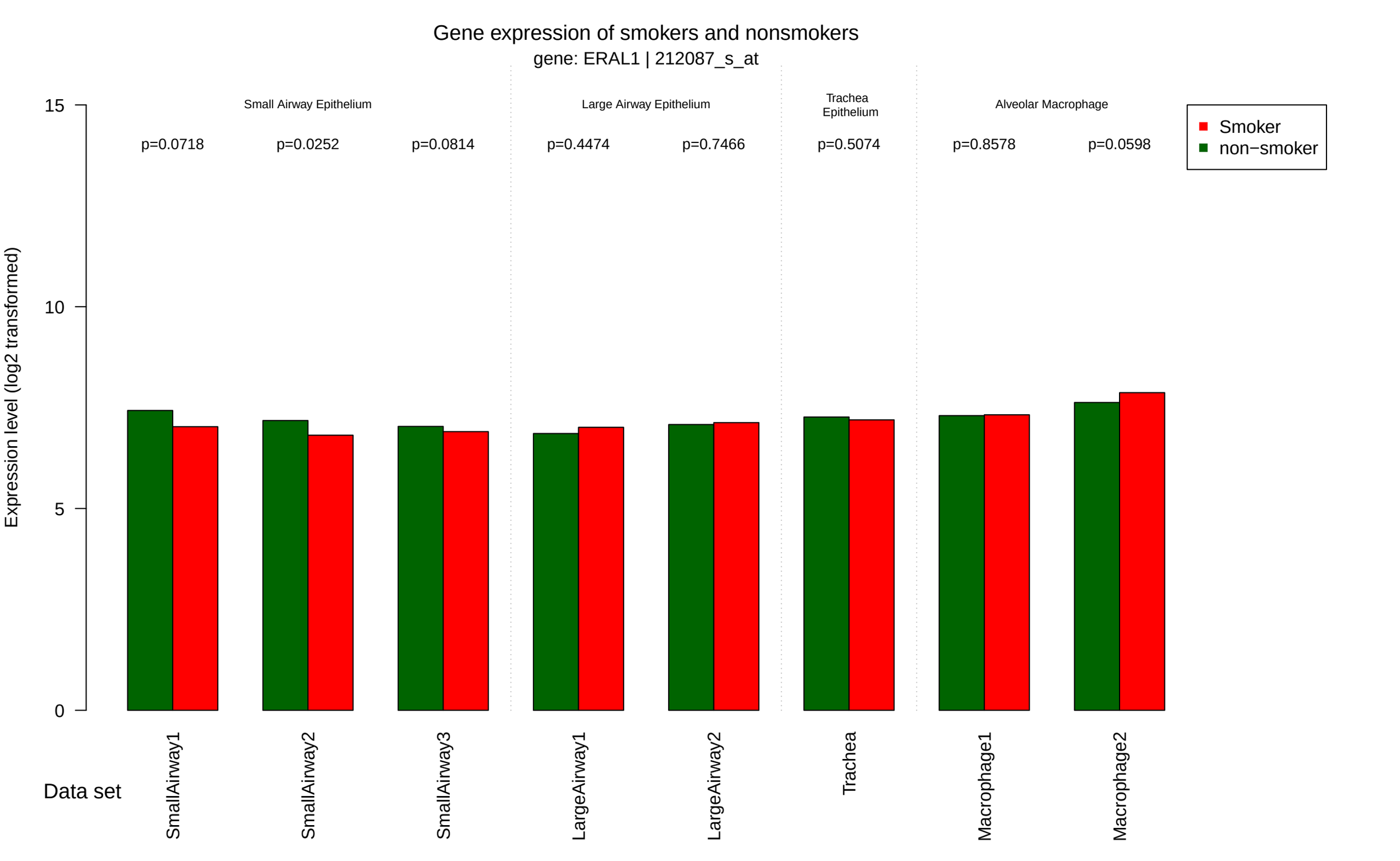
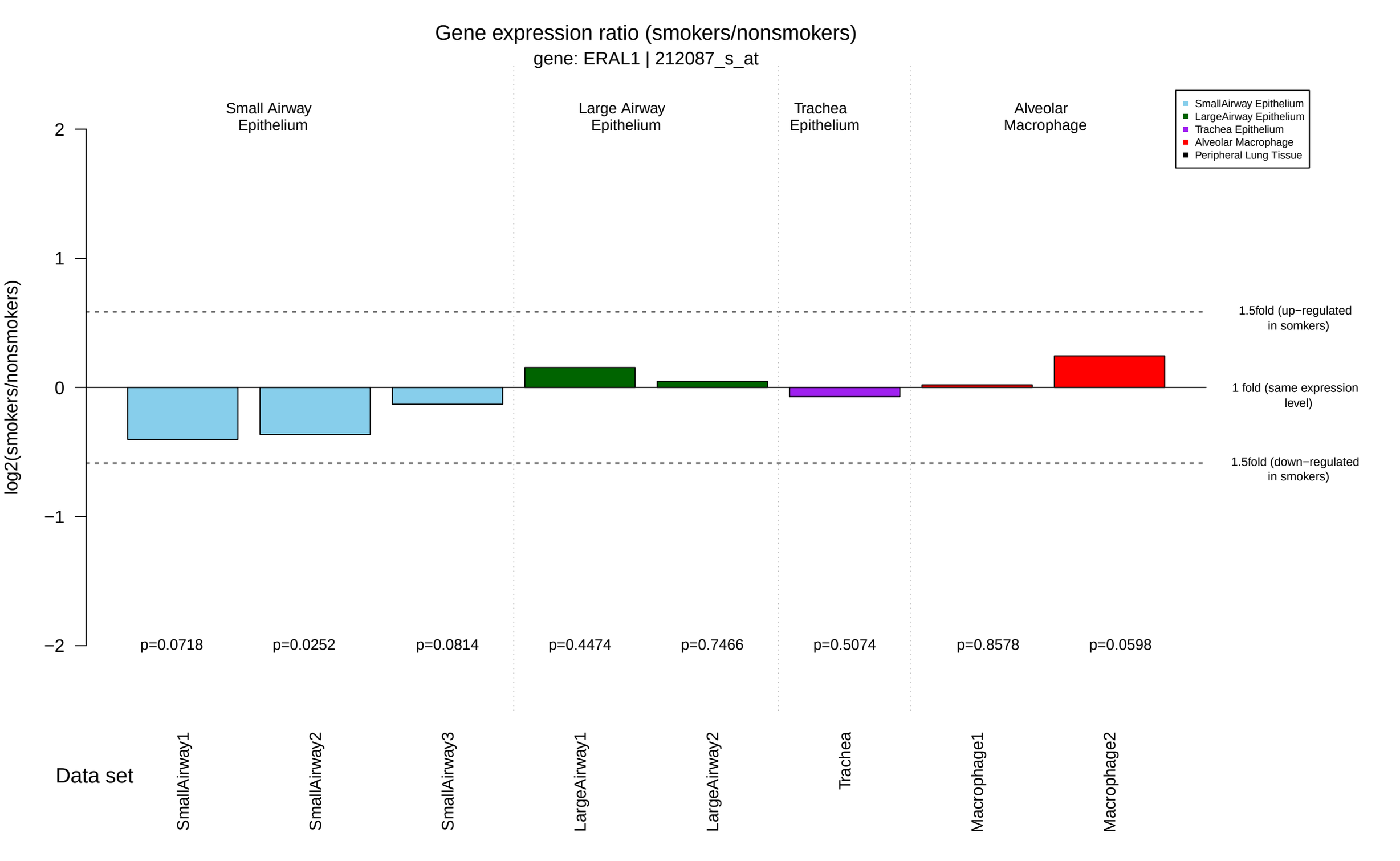
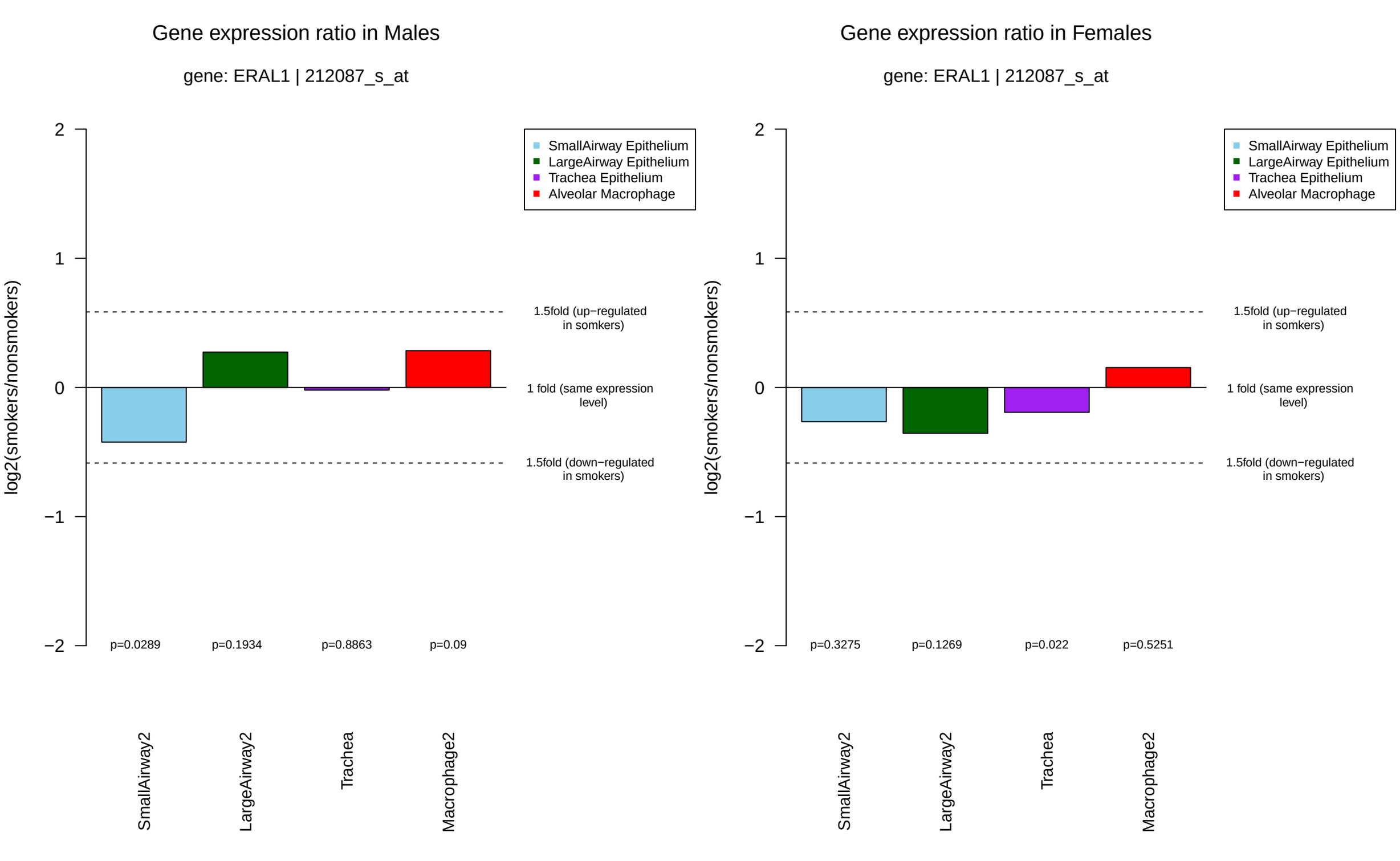
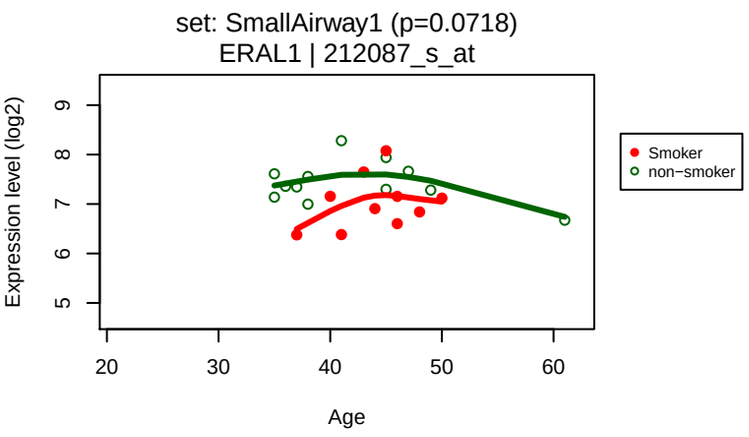
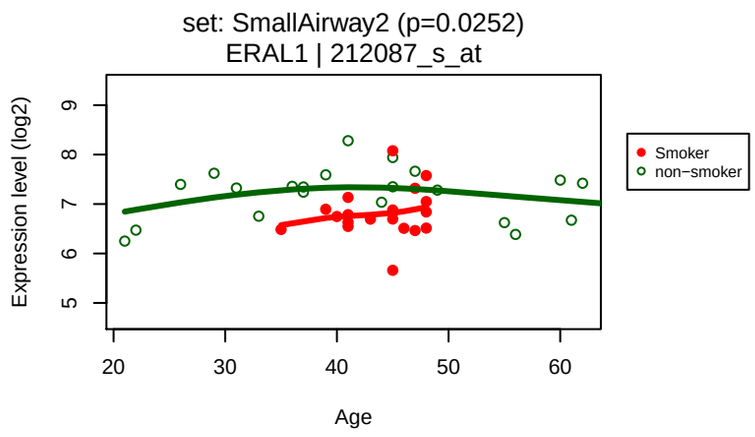
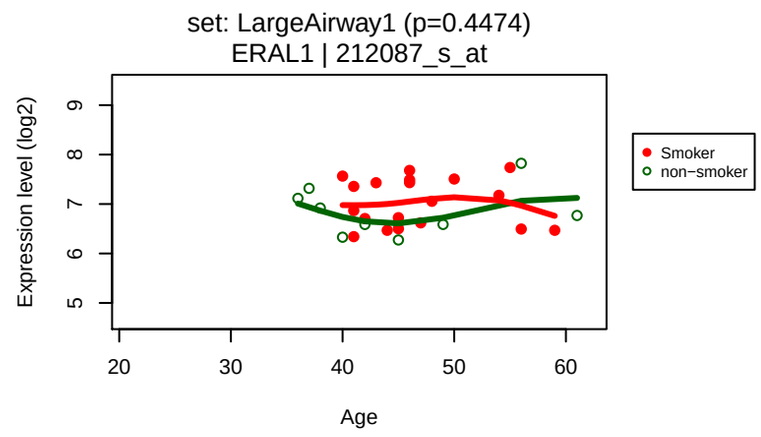

Gene expression of smokers and nonsmokers
gene: ERAL1 | 212087_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0718
p=0.0252
p=0.0814
p=0.4474
p=0.7466
p=0.5074
p=0.8578
p=0.0598
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

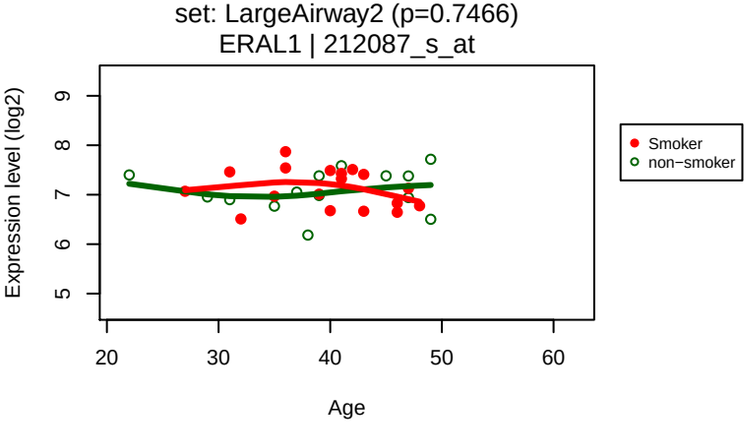

Gene expression ratio (smokers/nonsmokers)
gene: ERAL1 | 212087_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0718
p=0.0252
p=0.0814
p=0.4474
p=0.7466
p=0.5074
p=0.8578
p=0.0598
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

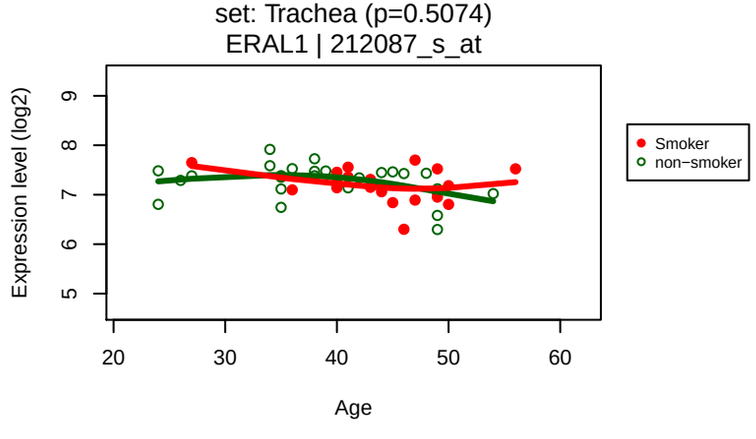

Gene expression ratio in Males
gene: ERAL1 | 212087_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0289
p=0.1934
p=0.8863
p=0.09
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ERAL1 | 212087_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3275
p=0.1269
p=0.022
p=0.5251
SmallAirway2
LargeAirway2
Trachea
Macrophage2

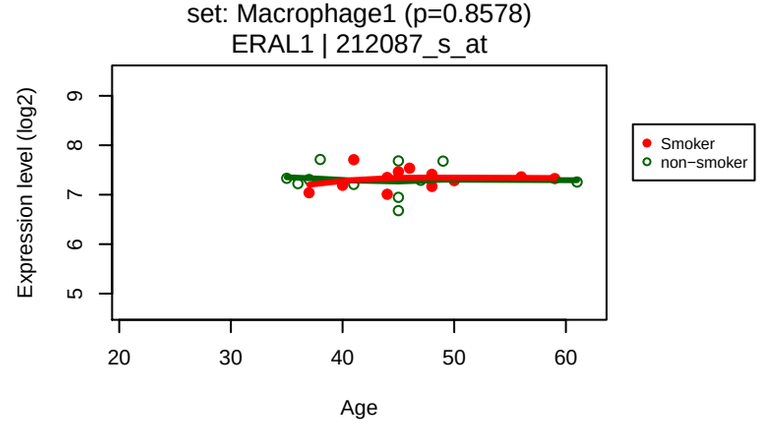

set: SmallAirway1 (p=0.0718)
ERAL1 | 212087_s_at
Expression level (log2)
Age
Smoker
non−smoker

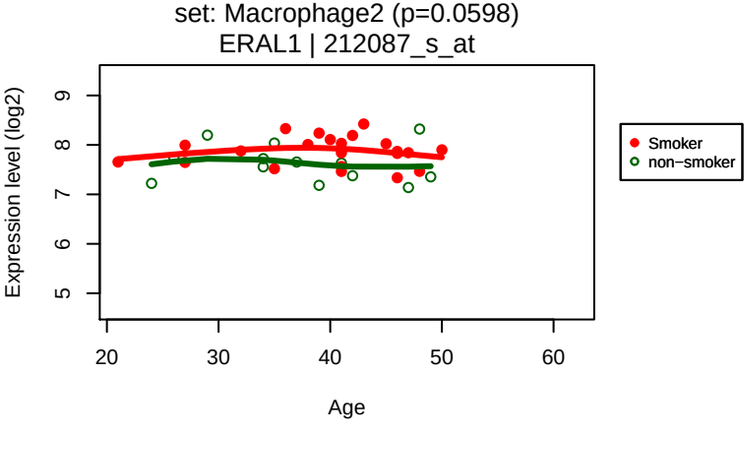

set: SmallAirway2 (p=0.0252)
ERAL1 | 212087_s_at
Expression level (log2)
Age
Smoker
non−smoker

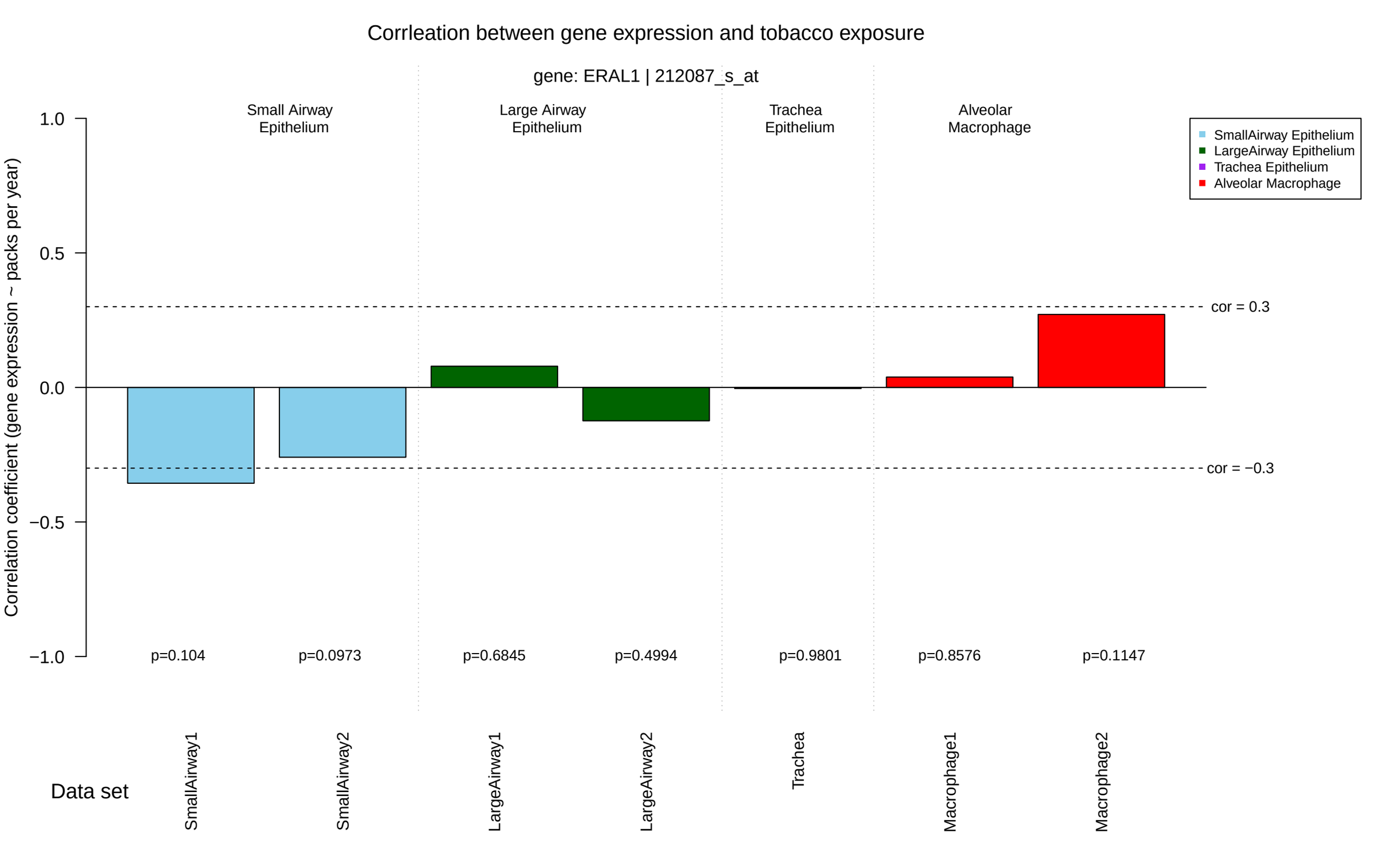

set: LargeAirway1 (p=0.4474)
ERAL1 | 212087_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7466)
ERAL1 | 212087_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5074)
ERAL1 | 212087_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8578)
ERAL1 | 212087_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0598)
ERAL1 | 212087_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ERAL1 | 212087_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.104
p=0.0973
p=0.6845
p=0.4994
p=0.9801
p=0.8576
p=0.1147
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2